

Advanced (Habanero)

Q1. Find the GCF of 12 and 18 using factors.

- (A) 1
- (B) 2
- (C) 3
- (D) 6
- (E) 12

Q2. Find the LCM of 9 and 15 using multiples.

- (A) 15
- (B) 30
- (C) 45
- (D) 60
- (E) 90

Q3. Find the GCF of 24 and 30 using prime factorization.

- (A) 1
- (B) 2
- (C) 3
- (D) 6
- (E) 12

Q4. A bookshelf has 12 shelves, and each shelf can hold 8 books. If the bookshelf is currently empty, what is the LCM of 12 and 8?

- (A) 12
- (B) 16
- (C) 24
- (D) 32
- (E) 48

Q5. Find the GCF of 50 and 75.

- (A) 1
- (B) 5
- (C) 10
- (D) 25
- (E) 50

Q6. A group of friends want to share some candy equally. If they have 48 pieces of candy and there are 8 friends, what is the LCM of 8 and 6?

- (A) 12
- (B) 16
- (C) 24
- (D) 32
- (E) 48

Q7. Find the GCF of 18 and 24 using prime factorization.

- (A) 1
- (B) 2
- (C) 3
- (D) 6
- (E) 12

Q8. A bakery sells 240 loaves of bread per day, and they come in packages of 12 or 15. What is the LCM of 12 and 15?

- (A) 30
- (B) 45
- (C) 60
- (D) 90
- (E) 120

Open Ended Questions (FRQ)

Q9. Tom has 18 boxes of pens, and each box contains 12 pens. If Alex has 24 boxes of pencils, and each box contains 15 pencils, what is the GCF and LCM of the total number of pens and pencils they have? Show your work.

Q10. A school is planning a field trip, and they have 3 buses with 15 seats each, and 2 buses with 20 seats each. What is the LCM of the total number of seats on each type of bus? If they want to fill the buses equally, what is the GCF of the total number of seats on each type of bus? Show your work.

Chef's Tips

- Make sure to check the work for question 9 and 10.
- Emphasize the importance of using prime factorization for GCF and LCM.
- Encourage students to use real-world examples to illustrate their understanding of GCF and LCM.